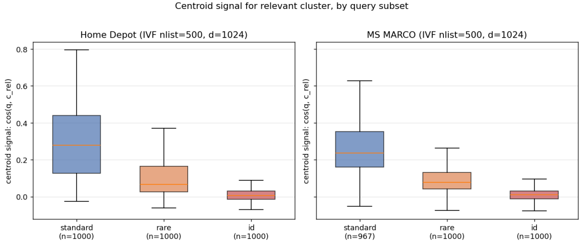
Direct evidence: measure the centroid signal

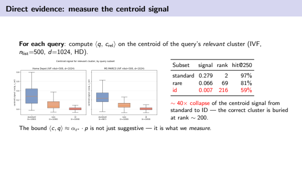
For each query: compute $\langle q, c_{\mathrm{rel}} \rangle$ on the centroid of the query's *relevant* cluster (IVF, n_{list}=500, d=1024, HD).



Subset	signal	rank	hit@250
standard	0.279	2	97%
rare	0.066	69	81%
id	0.007	216	59%

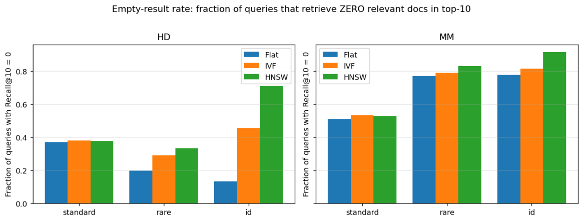
$\sim 40\times$ collapse of the centroid signal from standard to ID — the correct cluster is buried at rank $\sim$ 200.

The bound $\langle c, q \rangle \approx \alpha_{t^*} \cdot p$ is not just suggestive — it is what we *measure*.



Framing. This is where we stop arguing from theory and start measuring the exact quantity the bound predicts.
What we did (per query). Train IVF on Home Depot ($n_{\mathrm{list}} = 500$, $d = 1024$). For every query, look up which cluster actually contains its relevant doc(s) — call that centroid c_{rel}. Then compute $\langle q, c_{\mathrm{rel}} \rangle$, and also *where* c_{rel} ranks among all 500 centroids scored against q, and whether it makes the top 250 ("hit@250" — the halfway mark).
Histogram. Standard queries: signal ~ 0.28, clearly separated from zero. Rare queries collapse toward zero (~ 0.07). ID queries pile up at ~ 0.007 — **40$\times$ smaller than standard**.
Table. Standard: correct centroid at rank 2 on average, hit@250 = 97% — IVF with $n_{\mathrm{probe}} = 5$ is basically always correct. Rare: rank 69, so you'd need $n_{\mathrm{probe}} \approx 70$ (huge). **ID: rank 216 out of 500 — worse than median.** hit@250 = 59% means **41% of ID queries have their relevant cluster in the bottom half** of the centroid ranking. Those are structurally unrecoverable short of scanning all clusters — which defeats IVF's purpose.
Tie to the bound. $\langle c, q \rangle \approx \alpha_{t^*} \cdot p$. Rare tokens: huge α (IDF), tiny p (prevalence). Product falls under the JL floor ϵ — exactly the collapse we measure.
Anticipated question. "Just use finer clusters?" It helps (rare/ID gain 2–3$\times$ more relatively than standard going n_{list} 200$\rightarrow$1000) but you're fighting p: shrinking clusters means more centroids to score,

The production-visible symptom



Fraction of queries with zero relevant docs in top-10:

- HD-id: Flat **13%** → HNSW **71%**
- MM-id: Flat **78%** → HNSW **92%**
- Standard: within ±2 pp on both datasets.

Cost to match Flat recall (efSearch, HD):

- standard: **2.7× faster** than Flat
- id: **40× slower** than Flat

ANN's speedup evaporates on the population that needs it most.

Beyond Representation: Index Structure Limitations in Dense Retrieval

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Thank you

Questions?

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